NIOS Class 10 (212) Science & Technology — 7-Day Fast-Track Study Plan

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Class 10 (212) Science & Technology — 7-Day Fast-Track Study Plan

Designed for Working Students | Target: 55+ to 70+ Marks out of 85

> **Daily Time Commitment**: 2 hours on weekdays (45 mins morning + 1 hr 15 mins evening), 3–4 hours on weekend.

The Working Student's Daily Rhythm

- **Morning Sprint (45 mins)**: Read the Formula & Concept cheat sheet for that day's module. Memorize 3 key definitions or diagrams.
- **Commute / Breaks (15 mins)**: Review chemical formulas or SI units on your phone.
- **Evening Session (75 mins)**: Write down 2 diagrams, solve 1 physics numerical, and review 10 objective MCQs.

7-Day Fast-Track Roadmap

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graph TD
    D1["Day 1: High-Yield Physics Optics & Eye (15 Marks)<br>Mirror/Lens Formula + Ray Diagra
    D2 --> D3["Day 3 (Weekend): Biology Life Processes & Heredity (18 Marks)<br>Heart + Nephr
    D3 --> D4["Day 4 (Weekend): Chemistry Salts & Carbon (16 Marks)<br>4 Key Salts + Metals/N
    D4 --> D5["Day 5: Mechanics, Gravitation & Periodic Table (12 Marks)<br>F=ma + Gravitatio
    D5 --> D6["Day 6: Ecology & Top 50 Objectives Blitz (42 Marks)<br>10% Law + Food Chain +
    D6 --> D7["Day 7: Final Rehearsal & Mock Paper<br>Practice 1 Mock Test + Quick Glance at
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Day 1: High-Yield Physics — Light & Human Eye (15 Marks)

- **Target Chapters**: L.15 (Light Energy)
- **Core Topics**:
- Reflection & Refraction: Snell's Law ($\frac{\sin i}{\sin r} = \mu$).
- Mirror Formula ($\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$) and Lens Formula ($\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$).
- Power of Lens: $P = \frac{1}{f \text{ (in metres)}}$ (Unit: Dioptre, D).
- Eye Defects: Myopia (corrected by concave lens) vs Hypermetropia (corrected by convex lens) with ray diagrams.
- **Practice**: Draw the two eye correction diagrams on paper!

Day 2: High-Yield Physics — Electricity & Magnetism (15 Marks)

- **Target Chapters**: L.16 (Electrical Energy) & L.17 (Magnetic Effect)
- **Core Topics**:
- Ohm's Law: $V = IR$. Circuit diagram for verification.
- Equivalent resistance: Series ($R_s = R_1 + R_2 + R_3$) vs Parallel ($\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$).
- Electric Power ($P = VI = I^2R = \frac{V^2}{R}$) and Commercial Energy billing ($1\ \mathrm{kWh} = 1\ \text{unit} = 3.6 \times 10^6\ \mathrm{J}$).
- Fleming's Left-Hand Rule (Electric Motor) & Domestic Circuit Safety (Earth wire, Fuse).
- **Practice**: Solve 2 numerical problems on equivalent resistance and energy cost.

Day 3 (Weekend): Biology Core — Life Processes & Heredity (18 Marks)

- **Target Chapters**: L.22 (Life Processes I), L.23 (Control), L.24 (Reproduction), L.25 (Heredity)
- **Core Topics**:
- Human Heart & Double Circulation (Why is oxygenated separated from deoxygenated blood?).
- Nephron: Ultrafiltration and urine formation.
- Aerobic vs Anaerobic respiration pathways.
- Mendel's Monohybrid Cross: $3:1$ ratio and Punnett square.
- Sex determination in humans: Father (XY) determines gender of child.
- **Practice**: Sketch the heart diagram and the Mendel $3:1$ cross chart.

Day 4 (Weekend): Chemistry Core — Salts, Metals & Carbon (16 Marks)

- **Target Chapters**: L.8 (Acids, Bases & Salts), L.27 (Metals & Non-metals), L.28 (Carbon Compounds)
- **Core Topics**:
- Preparation and 2 uses of: Bleaching Powder, Baking Soda, Washing Soda, Plaster of Paris.
- pH scale: Acids (< 7), Neutral (7), Bases (> 7).
- Reactivity series & displacement reactions.
- Carbon: Catenation, Tetravalency, Homologous series, and Cleansing action of soap (Micelles).
- **Practice**: Write down the 4 salt preparation chemical equations twice.

Day 5: Mechanics, Gravitation & Periodic Table (12 Marks)

- **Target Chapters**: L.10 (Force & Motion), L.11 (Gravitation), L.5 & L.6 (Atoms & Periodic Table)
- **Core Topics**:
- Newton's Second Law: $F = ma$ and momentum ($p = mv$).
- Universal Law of Gravitation: $F = G\frac{Mm}{r^2}$. Difference between g (acceleration due to gravity) and G (Universal constant).
- Mass (constant, scalar) vs Weight ($W = mg$, variable, vector).
- Modern Periodic Table: Periodic trends across periods (left to right) and down groups (atomic size, valency, metallic character).

Day 6: Ecology & Top 50 Objectives Blitz (42 Marks!)

- **Target Chapters**: L.29 (Natural Environment), L.30 (Human Impact), and all objective questions.
- **Core Topics**:
- 10% Law of Energy Transfer (Lindeman's rule) in food chain.
- Ozone layer depletion: Reaction of Chlorofluorocarbons (CFCs).
- Biomagnification of pesticides (DDT).
- Solve the entire file: `03_TOP_50_REPEATED_OBJECTIVES_MCQ_SOLVED.md`.

Day 7: Final Rehearsal & Exam Day Warm-Up

- **Activities**:
- Attempt the 85-mark `06_PREDICTED_MOCK_TEST_PAPER_2026.md`.
- Check your answers against the provided solutions.

- Read the `02_ALL_IN_ONE_SCIENCE_FORMULA_AND_CONCEPTS_SHEET.md` on exam morning.
- Exam Day Kit: Admit card, black/blue ballpoint pens, pencil, eraser, ruler (for ray diagrams & circuits).

NIOS Class 10 (212) Science & Technology — Syllabus Bifurcation & Exam Blueprint

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Class 10 (212) Science & Technology — Syllabus Bifurcation & Exam Blueprint

> **CRITICAL TIME-SAVING RULE FOR WORKING STUDENTS:**
> NIOS has officially bifurcated the syllabus into **TMA (40%)** and **Public Examination (60%)**.
> Out of 32 total textbook chapters, **13 chapters are TMA-only and will NEVER appear in the written exam!**
> You only need to study the **19 Public Examination chapters**.

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1. What to Study vs What to Completely Skip

Module & Subject Area	Total	❌ DO NOT STUDY (TMA Only - 40%)	✅ MUST STUDY (Public Exam - 60%)
Module 1: Measurement	1	*None*	• **L.1: Measurement in Science & Technology**
Module 2: Matter (Chemistry)	7	• L.2: Matter in our Surroundings • L.3: Atom and Molecules • L.4: Chemical Reaction & Equations	• **L.5: Atomic Structure** • **L.6: Periodic Classification of Elements** • **L.7: Chemical Bonding** • **L.8: Acids, Bases and Salts**
Module 3: Moving Things (Physics)	3	• L.9: Motion and its Description	• **L.10: Force and Motion** • **L.11: Gravitation**
Module 4: Energy (Physics)	7	• L.12: Sources of Energy • L.14: Thermal Energy • L.18: Sound and Communication	• **L.13: Work and Energy** • **L.15: Light Energy (Optics)** • **L.16: Electrical Energy** • **L.17: Magnetic Effect of Electric Current**
Module 5: The Living World (Biology)	7	• L.19: Classification of Living Organisms • L.20: History of Life on Earth • L.21: Building Blocks of Life – Cell & Tissues	• **L.22: Life Processes – I (Nutrition, Respiration, Circulation, Excretion)** • **L.23: Life Processes – II (Control & Coordination)** • **L.24: Life Processes – III (Reproduction)** • **L.25: Heredity**
Module 6: Natural Resources (Chemistry)	3	• L.26: Air and Water	• **L.27: Metals and Non-metals** • **L.28: Carbon and Its Compounds**
Module 7: Human & Environment (Ecology)	4	• L.31: Food Production • L.32: Health and Hygiene	• **L.29: Natural Environment** • **L.30: Human Impact on Environment**

Module & Subject Area	Total	❌ DO NOT STUDY (TMA Only - 40%)	✅ MUST STUDY (Public Exam - 60%)
TOTAL	**32 Lessons**	**13 Lessons SKIPPED (Save 40% Study Time!)**	**19 High-Yield Lessons ONLY**

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2. Question Paper Pattern (85 Marks, 2½ Hours)

The written paper contains **43 questions** totaling **85 Marks** (+15 marks practical = 100 marks total):

Section A: Objective Questions — 42 Marks (49.4% of the Paper!)

- **Q1 to Q17 (17 Questions × 1 Mark = 17 Marks)**: Multiple Choice Questions (MCQs). Formula definitions, SI units, symbols, basic concepts.
- **Q18 to Q27 (10 Questions × 2 Marks = 20 Marks)**: Objective Type (each question has two 1-mark subparts: Fill in the blanks, True/False, Match the column).
- **Q28 (1 Question × 5 Marks = 5 Marks)**: Context / Case Study question with five 1-mark subparts.

> **Scoring Insight**: Getting 35+ on these 42 objective marks guarantees you pass easily without breaking a sweat!

Section B: Subjective Questions — 43 Marks

- **Q29 to Q37 (9 Questions × 2 Marks = 18 Marks)**: Very Short Answer (VSA). Definitions, 2-line reasoning, short calculations.
- **Q38 to Q42 (5 Questions × 3 Marks = 15 Marks)**: Short Answer (SA). Chemical reactions, ray diagrams, 3 differences.
- **Q43 (1 Question × 5 Marks = 5 Marks)**: Long Answer (LA) with internal choice (Human physiology or Physics laws/numerical).
- **Q44 (1 Question × 5 Marks = 5 Marks)**: Long Answer (LA) with internal choice (Electricity/Light or Carbon/Salts).

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3. The 35-Mark "Sure-Shot" Topic List

If you master these 7 recurring topics, you will effortlessly score 35+ marks in Section B alone: 1. **Light & Optics (5 Marks)**: Mirror/Lens formula numerical ($\frac{1}{f} = \frac{1}{v} \pm \frac{1}{u}$) and Myopia/Hypermetropia defects with ray diagrams. 2. **Electricity (5 Marks)**: Ohm's law ($V = IR$), series vs parallel equivalent resistance, and electric power/energy billing numerical. 3. **Chemical Salts (4 Marks)**: Preparation, chemical equations, and uses of Bleaching Powder, Baking Soda, Washing Soda, and Plaster of Paris. 4. **Carbon Compounds (4 Marks)**: Saponification, cleansing action of soaps (micelles), and functional groups. 5. **Life Processes (6 Marks)**: Nephron structure & function, Double circulation in human heart, and Aerobic vs Anaerobic respiration pathways. 6. **Heredity (4 Marks)**: Mendel's monohybrid cross ($3:1$ phenotypic ratio) and sex determination in humans (XX and XY). 7. **Environment (4 Marks)**: 10% law of energy flow, Food chain vs food web, and Ozone layer depletion.

NIOS Class 10 (212) Science & Technology — Master Formula & Concept Cheat Sheet

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Class 10 (212) Science & Technology — Master Formula & Concept Cheat Sheet

> **Keep this file open on your phone or print it out.**
> Over 65% of exam marks come from direct application of these formulas, chemical equations, and biological definitions.

PART 1: PHYSICS FORMULAS & SI UNITS

1. Motion, Force & Gravitation

- **Newton's Second Law**:

$$F = ma = \frac{m(v-u)}{t}$$

(Force F in Newtons N, mass m in kg, acceleration a in $\mathrm{m/s^2}$)

- **Momentum**:

$$p = mv \quad (\text{SI Unit: } \mathrm{kg \cdot m/s})$$

- **Universal Law of Gravitation**:

$$F = G\frac{m_1 m_2}{r^2}$$

($G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$ is the Universal Gravitational Constant)

- **Acceleration due to gravity (g)**:

$$g = \frac{GM}{R^2} \approx 9.8 \text{ m/s}^2 \quad (\text{On Moon, } g_{\text{moon}} = \frac{1}{6} g_{\text{earth}})$$

- **Weight**:

$$W = mg \quad (\text{Weight is a force, measured in Newtons N})$$

- **Archimedes' Principle**: $\text{Buoyant Force} = \text{Weight of fluid displaced.}$

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2. Work, Energy & Power

- **Work Done**:

$$W = F \times s \quad (\text{SI Unit: Joule, J})$$

(1 Joule is the work done when a force of 1 N moves a body through a distance of 1 m)

- **Kinetic Energy**:

$$KE = \frac{1}{2}mv^2$$

- **Gravitational Potential Energy**:

$$PE = mgh$$

- **Power**:

$$P = \frac{\text{Work}}{\text{Time}} = \frac{W}{t} \quad (\text{SI Unit: Watt, W})$$

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3. Light (Optics)

- **Snell's Law of Refraction**:

$$\mu = \frac{\sin i}{\sin r} = \text{Constant}$$

- **Mirror Formula (Spherical Mirrors)**:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

- Magnification: $m = -\frac{v}{u} = \frac{h_i}{h_o}$

- **Lens Formula (Spherical Lenses)**:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

- Magnification: $m = +\frac{v}{u} = \frac{h_i}{h_o}$
- **Sign Convention (Cartesian)**:
- Object distance u is ALWAYS negative $(-)$.
- Concave mirror / Concave lens: focal length f is ALWAYS negative $(-)$.
- Convex mirror / Convex lens: focal length f is ALWAYS positive $(+)$.
- **Power of a Lens**:

$$P = \frac{1}{f \text{ (in metres)}} \quad (\text{SI Unit: Dioptre, D})$$

- Convex lens $\implies$ Positive power (e.g. $+2$ D).
- Concave lens $\implies$ Negative power (e.g. -1.5 D).
- **Human Eye Defects & Correction**:
- **Myopia (Short-sightedness)**: Can see near objects clearly, cannot see distant objects. Image forms in front of retina. Corrected using **Concave Lens** of suitable power.
- **Hypermetropia (Long-sightedness)**: Can see distant objects clearly, cannot see near objects. Image forms behind retina. Corrected using **Convex Lens** of suitable power.

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4. Electricity & Magnetism

- **Electric Current**:

$$I = \frac{Q}{t} \quad (\text{Current in Amperes A}, \ Q \text{ in Coulombs C})$$

- **Potential Difference**:

$$V = \frac{W}{Q} \quad (\text{Voltage in Volts V}, \ W \text{ in Joules J})$$

- **Ohm's Law**:

$$V = IR \implies R = \frac{V}{I} \quad (\text{Resistance in Ohms}, \Omega)$$

- **Resistance & Resistivity**:

$$R = \rho \frac{l}{A}$$

(l = length in m, A = cross-sectional area in m^2, ρ = resistivity in $\Omega \cdot \mathrm{m}$)

- **Resistors in Series**:

$$R_s = R_1 + R_2 + R_3 \quad (\text{Current } I \text{ remains same, voltage divides})$$

- **Resistors in Parallel**:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad (\text{Voltage } V \text{ remains same, current divides})$$

- **Joule's Law of Heating**:

$$H = I^2 R t = V I t = \frac{V^2}{R} t \quad (\text{in Joules})$$

- **Electric Power**:

$$P = V I = I^2 R = \frac{V^2}{R} \quad (\text{in Watts})$$

- **Commercial Unit of Electrical Energy**:

$$1\ \mathrm{kWh} = 1\ \mathrm{Unit} = 1000\ \mathrm{W} \times 3600\ \mathrm{s} = 3.6 \times 10^6\ \mathrm{Joules}$$

- **Domestic Circuit Wires**:
- Live Wire: Red or Brown (carries current at $220\ \mathrm{V}$)
- Neutral Wire: Black or Light Blue (completes circuit at $0\ \mathrm{V}$)
- Earth Wire: Green or Yellow (safety measure connected to metal casing)
- **Fleming's Left-Hand Rule (Electric Motor)**:
- **Thumb**: Direction of Force / Motion
- **Forefinger**: Direction of Magnetic Field
- **Middle Finger**: Direction of Current

PART 2: CHEMISTRY FORMULAS, EQUATIONS & SALTS

1. The 4 Essential Chemical Salts (Guaranteed 4 to 6 Marks!)

Chemical Name	Common Name	Chemical Formula	Preparation Equation	Key Practical Uses
Calcium Oxychloride	Bleaching Powder	$CaOCl_2$	$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$	1. Disinfectant in drinking water 2. Bleaching cotton in textiles
Sodium Hydrogen Carbonate	Baking Soda	$NaHCO_3$	$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$	1. Ingredient in baking powder 2. Antacid for acidity relief
Sodium Carbonate Decahydrate	Washing Soda	$Na_2CO_3 \cdot 10H_2O$	$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O$	1. Softening permanent hard water 2. Glass, soap, and paper industries
Calcium Sulphate Hemihydrate	Plaster of Paris (POP)	$CaSO_4 \cdot \frac{1}{2}H_2O$	$CaSO_4 \cdot 2H_2O \xrightarrow{373\ K} CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O$	1. Plastering fractured bones 2. Making statues and decorative casts

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2. Acids, Bases & pH Indicators

- **pH Scale**:
- $pH < 7 \implies$ Acidic (e.g. Gastric juice $pH \approx 1.2$, Lemon juice ≈ 2.2)
- $pH = 7 \implies$ Neutral (Pure water, $NaCl$ solution)
- $pH > 7 \implies$ Basic / Alkaline (Blood ≈ 7.4, Milk of Magnesia ≈ 10, $NaOH \approx 14$)
- **Indicator Color Changes**:
- **Blue Litmus**: Turns RED in acid; remains blue in base.
- **Red Litmus**: Turns BLUE in base; remains red in acid.
- **Phenolphthalein**: Colourless in acid; turns PINK in base.

- **Methyl Orange**: RED in acid; YELLOW in base.

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3. Metals & Metallurgy

- **Reactivity Series**:

$$\mathrm{K} > \mathrm{Na} > \mathrm{Ca} > \mathrm{Mg} > \mathrm{Al} > \mathrm{Zn} > \mathrm{Fe} > \mathrm{Pb} > [\mathrm{H}] > \mathrm{Cu} > \mathrm{Hg} > \mathrm{Ag} > \mathrm{Au}$$

  (Potassium and Sodium are stored under kerosene oil because they react vigorously with atmospheric oxygen and moisture)
- **Roasting vs Calcination**:
- **Roasting**: Heating sulphide ore in **excess air** below its melting point:

$$2\mathrm{ZnS} + 3\mathrm{O_2} \xrightarrow{\Delta} 2\mathrm{ZnO} + 2\mathrm{SO_2}$$

- **Calcination**: Heating carbonate ore in **absence or limited air**:

$$\mathrm{ZnCO_3} \xrightarrow{\Delta} \mathrm{ZnO} + \mathrm{CO_2}$$

- **Thermite Reaction**:

$$\mathrm{Fe_2O_3} + 2\mathrm{Al} \rightarrow 2\mathrm{Fe\ (molten)} + \mathrm{Al_2O_3} + \mathrm{Heat} \quad \text{(Used to weld railway tracks)}$$

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4. Carbon and Its Compounds

- **Why Carbon forms millions of compounds**:
- **Catenation**: Unique property of carbon atoms to form long chains with other carbon atoms.
- **Tetravalency**: Valency of 4; forms strong covalent bonds with 4 other atoms.
- **Homologous Series**:
- **Alkanes** (single bond): C_nH_{2n+2} (Methane $\mathrm{CH_4}$, Ethane $\mathrm{C_2H_6}$, Propane $\mathrm{C_3H_8}$)
- **Alkenes** (double bond): C_nH_{2n} (Ethene $\mathrm{C_2H_4}$)
- **Alkynes** (triple bond): C_nH_{2n-2} (Ethyne / Acetylene $\mathrm{C_2H_2}$)
- **Functional Groups**:
- Alcohol: $-\mathrm{OH}$ (Ethanol $\mathrm{C_2H_5OH}$)
- Aldehyde: $-\mathrm{CHO}$ (Ethanal $\mathrm{CH_3CHO}$)
- Ketone: $>\mathrm{C{=}O}$ (Propanone / Acetone $\mathrm{CH_3COCH_3}$)
- Carboxylic Acid: $-\mathrm{COOH}$ (Ethanoic / Acetic Acid $\mathrm{CH_3COOH}$)

- **Saponification (Soap Making)**:

$$\text{Ester (Fat/Oil)} + \text{NaOH} \rightarrow \text{Soap (Sodium salt of fatty acid)} + \text{Glycerol}$$

- **Soap Cleansing Action**:
- Soap molecule has two ends:
  - **Hydrophilic ionic head**: Attracted to water (points outward).
  - **Hydrophobic hydrocarbon tail**: Attracted to oil/dirt (points inward).
- Together they form spherical clusters called **Micelles**, trapping oil droplets to be rinsed away with water.

PART 3: BIOLOGY CORE CONCEPTS & GENETICS

1. Life Processes & Physiology

- **Photosynthesis Equation**:

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

- **Breakdown of Glucose by Various Pathways**:

$$\text{Glucose (6C)} \xrightarrow{\text{in Cytoplasm}} \text{Pyruvate (3C)} + \text{Energy}$$

- (a) **In absence of O_2 (Yeast / Fermentation)**: $\rightarrow \text{Ethanol} + CO_2 + \text{Energy (2 ATP)}$
- (b) **In lack of O_2 (Muscle cells)**: $\rightarrow \text{Lactic acid} + \text{Energy (Causes muscle cramps)}$
- (c) **In presence of O_2 (Mitochondria / Aerobic)**: $\rightarrow CO_2 + H_2O + \text{Energy (38 ATP)}$
- **Double Circulation in Human Heart**:
- Blood passes through the heart **twice** in one complete cardiac cycle:
  1. **Pulmonary Circulation**: Deoxygenated blood from right ventricle $\rightarrow$ Lungs $\rightarrow$ Oxygenated blood to left atrium.
  2. **Systemic Circulation**: Oxygenated blood from left ventricle $\rightarrow$ Body organs $\rightarrow$ Deoxygenated blood to right atrium.
- **Nephron (Excretory Unit of Kidney)**:
- Composed of: Bowman's capsule, Glomerulus, Renal tubule, Collecting duct.
- Three steps of urine formation:
  1. **Ultrafiltration** (Glomerular filtration)
  2. **Selective reabsorption** (Glucose, amino acids, salts, water)
  3. **Tubular secretion**.

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2. Genetics & Heredity

- **Mendel's Monohybrid Cross (Tall $\times$ Dwarf Pea Plant)**:
- Parents: $TT \times tt$
- F_1 Generation: All Tt (All Tall)
- F_2 Generation:
  - **Phenotypic Ratio**: $\mathbf{3 : 1}$ (3 Tall : 1 Dwarf)
  - **Genotypic Ratio**: $\mathbf{1 : 2 : 1}$ ($1\,TT : 2\,Tt : 1\,tt$)
- **Sex Determination in Humans**:
- Female gametes (Eggs) carry only X chromosome ($22 + X$).
- Male gametes (Sperms) carry either X or Y chromosome ($22 + X$ or $22 + Y$).
- If sperm carrying X fertilizes egg $\rightarrow XX \implies$ **Girl Child**.
- If sperm carrying Y fertilizes egg $\rightarrow XY \implies$ **Boy Child**.
- **Conclusion**: The father's sperm determines the sex of the baby with a 50% probability for each gender.

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3. Ecology & Environment

- **10% Law of Energy Transfer (Lindeman's Rule)**:
- Only 10% of energy is transferred from one trophic level to the next.
- 90% of energy is lost as heat to the environment during metabolic processes.
- *Example*: Sun ($10,000$ J) $\rightarrow$ Plants (100 J) $\rightarrow$ Deer (10 J) $\rightarrow$ Tiger (1 J).
- **Ozone Layer Depletion**:
- Ozone (O_3) in stratosphere shields Earth from harmful UV radiation.
- Depleted by **Chlorofluorocarbons (CFCs)** from refrigerators and fire extinguishers.

$$CF_2Cl_2 \xrightarrow{UV} CF_2Cl + Cl^{\bullet}$$

$$Cl^{\bullet} + O_3 \rightarrow ClO^{\bullet} + O_2$$

- **Biomagnification**:
- Increase in concentration of non-biodegradable toxic substances (like DDT, mercury) at progressively higher trophic levels of a food chain. Highest concentration accumulates in top carnivores / humans.

NIOS Class 10 (212) Science & Technology — Top 50 Most Repeated Objective Questions

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Class 10 (212) Science & Technology — Top 50 Most Repeated Objective Questions

Master the 42 Objective Marks in Section A (MCQs, True/False & Blanks)

> **Section A carries 42 marks out of 85.**
> Scoring 35+ on these questions guarantees a passing grade before you even write a single descriptive answer in Section B!

Part 1: Multiple Choice Questions (1 Mark Each)

1. Physics (Optics, Mechanics & Electricity)

Q1. The SI unit of electric potential difference is:
(A) Ampere
(B) Volt
(C) Ohm
(D) Watt

> **Answer: (B) Volt**
> *Explanation*: $V = W/Q$ (Joules per Coulomb = Volt).

Q2. The commercial unit of electrical energy is kilowatt-hour (kWh). $1\ \mathrm{kWh}$ is equal to:
(A) $3.6 \times 10^5\ \mathrm{J}$
(B) $3.6 \times 10^6\ \mathrm{J}$

(C) 36×10^6 J
(D) 3.6×10^4 J

> **Answer: (B)** 3.6×10^6 J
> *Explanation*: $1\ \text{kW} \times 1\ \text{hr} = 1000\ \text{W} \times 3600\ \text{s} = 3.6 \times 10^6\ \text{J}$.

Q3. A person cannot see distant objects clearly, but can see nearby objects distinctly. The defect of vision he is suffering from is:
(A) Hypermetropia
(B) Myopia
(C) Presbyopia
(D) Cataract

> **Answer: (B) Myopia**
> *Explanation*: Myopia is short-sightedness, corrected with a concave lens.

Q4. A lens has a power of $+2.0\ \text{D}$. Its focal length is:
(A) $+50\ \text{cm}$
(B) $-50\ \text{cm}$
(C) $+20\ \text{cm}$
(D) $+100\ \text{cm}$

> **Answer: (A) +50 cm**
> *Explanation*: $f = \frac{1}{P} = \frac{1}{+2} = +0.5\ \text{m} = +50\ \text{cm}$ (Convex lens).

Q5. If the resistance of a wire of length l is R, then another wire of the same material having double the length and same cross-sectional area will have resistance:
(A) $R/2$
(B) $2R$
(C) $4R$
(D) R

> **Answer: (B) 2R**
> *Explanation*: $R \propto l$. Doubling length doubles the resistance.

Q6. The acceleration due to gravity on the surface of the Moon is approximately:
(A) $\frac{1}{2}$ that of Earth
(B) $\frac{1}{6}$ that of Earth
(C) 6 times that of Earth
(D) Equal to that of Earth

> **Answer: (B) 1/6 that of Earth**

Q7. An object of mass $10\ \text{kg}$ is taken to the Moon. Its mass on the Moon will be:
(A) $10\ \text{kg}$
(B) $\frac{10}{6}\ \text{kg}$
(C) $60\ \text{kg}$
(D) $0\ \text{kg}$

> **Answer: (A) 10 kg**
> *Explanation*: Mass is constant everywhere. (Weight changes to $\frac{1}{6}$, but mass remains $10\ \text{kg}$).

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2. Chemistry (Acids, Salts, Metals & Carbon)

Q8. The chemical formula of Bleaching Powder is:
(A) CaCl_2
(B) CaOCl_2
(C) Ca(OH)_2
(D) CaSO_4

> **Answer: (B)** CaOCl_2 (Calcium oxychloride)

Q9. Which of the following salts is used for supporting fractured bones in medicine?
(A) Bleaching powder
(B) Baking soda
(C) Plaster of Paris
(D) Washing soda

> **Answer: (C) Plaster of Paris** $(\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O})$

Q10. The pH of an aqueous solution is measured to be 3. The solution is:
(A) Strongly alkaline
(B) Weakly alkaline
(C) Strongly acidic
(D) Neutral

> **Answer: (C) Strongly acidic** (pH < 7 is acidic; lower pH means higher acidity).

Q11. A metal which is liquid at room temperature is:
(A) Sodium
(B) Mercury
(C) Bromine
(D) Gallium

> **Answer: (B) Mercury (Hg)**
> *(Note: Bromine is a liquid NON-metal; Mercury is a liquid METAL)*.

Q12. Which of the following metals is stored under kerosene oil to prevent accidental fire?
(A) Magnesium
(B) Aluminium
(C) Sodium
(D) Iron

> **Answer: (C) Sodium (Na)** (and Potassium).

Q13. The non-metal which is a good conductor of electricity is:
(A) Diamond
(B) Graphite
(C) Sulphur
(D) Phosphorus

> **Answer: (B) Graphite** (Allotrope of carbon with free delocalized electrons).

Q14. The functional group present in an alcohol molecule is:
(A) $-\mathrm{CHO}$
(B) $-\mathrm{COOH}$

(C) $-\mathrm{OH}$
(D) $>\mathrm{C}=\mathrm{O}$

> **Answer: (C) -OH**

---

3. Biology & Environment

Q15. The basic structural and functional filtration unit of the human kidney is:
(A) Neuron
(B) Nephron
(C) Alveolus
(D) Glomerulus

> **Answer: (B) Nephron**

Q16. In humans, fertilization of the egg by a sperm normally takes place in the:
(A) Uterus
(B) Ovary
(C) Fallopian tube (Oviduct)
(D) Vagina

> **Answer: (C) Fallopian tube (Oviduct)**

Q17. According to Lindeman's 10% law, if $1000\ \mathrm{J}$ of solar energy is trapped by producers, the energy available to secondary consumers will be:
(A) $100\ \mathrm{J}$
(B) $10\ \mathrm{J}$
(C) $1\ \mathrm{J}$
(D) $0.1\ \mathrm{J}$

> **Answer: (B) 10 J**
> *Explanation*: Producers $(1000\ \mathrm{J}) \rightarrow$ Herbivores $(100\ \mathrm{J}) \rightarrow$ Carnivores $(10\ \mathrm{J})$.

Part 2: Fill in the Blanks (2 Marks Each / 1 Mark Per Part)

Q18.

(i) Newton's first law of motion is also known as the law of _____.

> **Answer: inertia**

(ii) The product of mass and velocity of a body is called _____.

> **Answer: momentum** ($p = mv$)

Q19.

(i) The device used to measure electric current in a circuit is called _____.

> **Answer: ammeter**

(ii) An ammeter is always connected in _____ in an electrical circuit.

> **Answer: series** (Voltmeter is connected in parallel)

Q20.

(i) The focal length of a plane mirror is _____.

> **Answer: infinity**

(ii) The power of a concave lens is always _____.

> **Answer: negative**

Q21.

(i) The chemical formula of Baking Soda is _____.

> **Answer:** $\mathrm{NaHCO_3}$ (Sodium hydrogen carbonate)

(ii) Heating Gypsum at $373\ \mathrm{K}$ produces _____.

Answer: Plaster of Paris ($\mathrm{CaSO_4 \cdot \frac{1}{2}H_2O}$)

Q22.

(i) The property of self-linking of carbon atoms through covalent bonds is known as _____.

Answer: catenation

(ii) The general formula of Alkanes is _____.

Answer: $\mathrm{C_nH_{2n+2}}$

Q23.

(i) The hormone secreted by the pancreas that regulates blood sugar level is _____.

Answer: insulin

(ii) The plant hormone responsible for phototropism (bending towards light) is _____.

Answer: auxin

Q24.

(i) In Mendel's monohybrid cross, the phenotypic ratio in the F_2 generation is _____.

Answer: $3:1$

(ii) Normal human somatic cells contain _____ pairs of chromosomes.

Answer: 23 (46 chromosomes total)

Q25.

(i) The gas responsible for depleting the stratospheric ozone layer is _____.

Answer: Chlorofluorocarbon (CFC)

(ii) The progressive accumulation of non-biodegradable toxic substances along a food chain is termed _____.

> **Answer: biomagnification**

Part 3: True / False Statements (2 Marks Each / 1 Mark Per Part)

Q26.

(i) "Weight of an object remains constant everywhere in the universe."

> **Answer: FALSE**
> *Reason*: Mass remains constant; weight depends on the local acceleration due to gravity ($W = mg$).

(ii) "Action and reaction forces act on two different bodies."

> **Answer: TRUE**
> *Reason*: Newton's Third Law states forces act on different bodies in opposite directions.

Q27.

(i) "The resistivity of a metallic wire depends on its length and thickness."

> **Answer: FALSE**
> *Reason*: Resistance (R) depends on length and area, but resistivity (ρ) is an intrinsic material property depending only on the nature of the material and temperature.

(ii) "An electric fuse wire should have high resistance and a low melting point."

> **Answer: TRUE**
> *Reason*: So that it melts quickly and breaks the circuit in case of an overload.

Q28.

(i) "A convex mirror always forms a virtual, erect, and diminished image."

> **Answer: TRUE**
> *Reason*: This is why convex mirrors are used as rear-view mirrors in vehicles.

(ii) "Red light bends the most when passing through a glass prism."

> **Answer: FALSE**
> *Reason*: Violet light bends the most; red light has the longest wavelength and bends the least.

Q29.
(i) "Pure water has a pH equal to 7 at $25^{\circ}\mathrm{C}$."

> **Answer: TRUE**

(ii) "Acids turn red litmus paper blue."

> **Answer: FALSE**
> *Reason*: Bases turn red litmus blue; acids turn blue litmus red.

Q30.
(i) "Saponification is the chemical reaction used to prepare soaps."

> **Answer: TRUE**

(ii) "Diamond and graphite are chemical isomers of carbon."

> **Answer: FALSE**
> *Reason*: They are allotropes, not isomers.

Q31.
(i) "Deoxygenated blood enters the left atrium of the human heart."

> **Answer: FALSE**
> *Reason*: Deoxygenated blood from the body enters the RIGHT atrium; oxygenated blood enters the LEFT atrium.

(ii) "In humans, the male gamete determines the sex of the unborn child."

> **Answer: TRUE**
> *Reason*: The father donates either an X or Y chromosome, whereas the mother always donates an X.

Q32.

(i) "Energy flow in an ecosystem is unidirectional."

> **Answer: TRUE**
> *Reason*: Energy flows from the Sun $\rightarrow$ producers $\rightarrow$ herbivores $\rightarrow$ carnivores and cannot flow backward.

(ii) "Biodegradable substances are the primary cause of biomagnification."

> **Answer: FALSE**
> *Reason*: Non-biodegradable chemicals like DDT cause biomagnification because they cannot be broken down.

NIOS Class 10 (212) Science & Technology — Guaranteed Sure-Shot Questions & Diagrams

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Class 10 (212) Science & Technology — Guaranteed Sure-Shot Questions & Diagrams

> **These 6 question templates appear with near-100% certainty in NIOS Science & Technology papers.**
> Mastering these 6 templates locks in **25 to 30 marks** in Section B! Practice drawing the diagrams and writing the equations with a pen.

Template 1: Human Eye Defects — Myopia vs Hypermetropia (5 Marks — Q43/Q44)

Problem:

> What is Myopia? State its two causes. How can this defect be corrected? Draw ray diagrams showing:
> (i) a myopic eye, and
> (ii) its correction using an appropriate lens.

Complete Model Answer:

1. **Definition**:
2. **Myopia (Short-sightedness)** is a defect of vision in which a person can see nearby objects distinctly, but cannot see distant objects clearly. The far point of a myopic person is nearer than infinity.

3. **Causes**:
4. Excessive curvature of the eye lens (lens becomes too thick / convergent).
5. Elongation of the eyeball.
6. **Nature of Image & Correction**:
7. Light rays coming from distant objects focus **in front of the retina** instead of on the retina.
8. **Correction**: It is corrected by using a **Concave Lens** (diverging lens) of suitable focal length. The concave lens diverges the parallel rays so that they appear to come from the person's far point, allowing the eye lens to focus the image exactly on the retina.
9. **Ray Diagrams (Representation)**:

```
(i) MYOPIC EYE (Image formed in FRONT of retina):
Parallel rays ---------\
                        \--\
----------------------------[ Eye Lens ]-----> (Focuses HERE) ----- [ Retina ]
                        /--/                   (Blurry on Retina)
Parallel rays ---------/

(ii) CORRECTION USING CONCAVE LENS:
Parallel rays -----\
                    [ Concave ] -------\
--------------------[  Lens   ]---------[ Eye Lens ]----------------> [ Retina ]
                    [ (Diverges)]------/                              (Sharp Focus!)
Parallel rays -----/
```

*(Note for Hypermetropia / Long-sightedness: Image forms BEHIND the retina due to eyeball being too short or lens being too thin. It is corrected using a **Convex Lens**).*

Template 2: Ohm's Law Verification & Equivalent Resistance (5 Marks — Q43)

Problem:

> (a) State Ohm's law. Draw a schematic circuit diagram to verify Ohm's law in the laboratory.
> (b) Three resistors of resistances $2\,\Omega$, $3\,\Omega$, and $6\,\Omega$ are connected. How can they be connected to give a total resistance of:
> (i) $4\,\Omega$, and (ii) $1\,\Omega$?

Complete Model Answer:

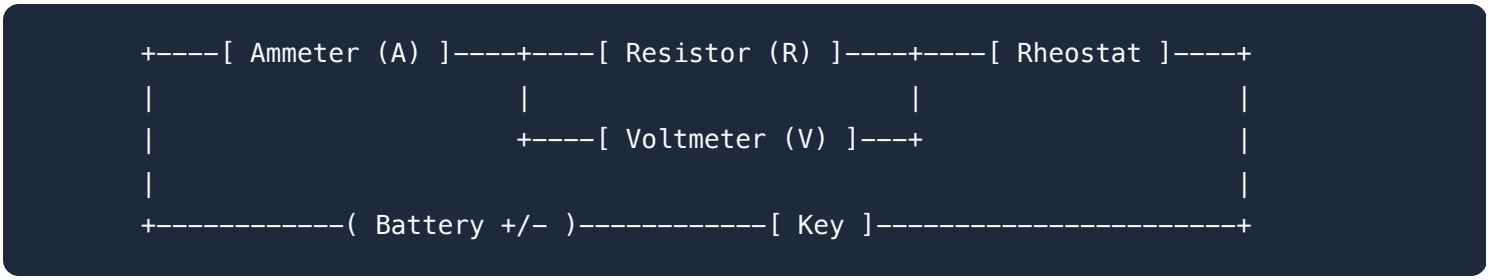
1. **Statement of Ohm's Law**:
2. Ohm's law states that the electric current (I) flowing through a metallic conductor is directly proportional to the potential difference (V) applied across its ends, provided temperature and other physical conditions remain constant:

$$V \propto I \implies V = IR$$

3. *(where R is the electrical resistance of the conductor)*.

4. **Circuit Diagram**:

5. Components connected in series: Battery (cell), Plug key, Ammeter (A), Rheostat (variable resistor).
6. Component connected in parallel: Voltmeter (V) connected across the resistor (R).



1. **Numerical Solution**:
2. **(i) To get $4\,\Omega$**:

- Connect $3\,\Omega$ and $6\,\Omega$ in **parallel**, and then connect that combination in **series** with $2\,\Omega$:

$$\frac{1}{R_p} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6} = \frac{1}{2} \implies R_p = 2\,\Omega$$

$$R_{\text{total}} = 2\,\Omega + R_p = 2\,\Omega + 2\,\Omega = \mathbf{4\,\Omega}.$$

3. **(ii) To get** $1\,\Omega$:
   - Connect all three resistors in **parallel**:

$$\frac{1}{R_p} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = \frac{6}{6} = 1 \implies R_{\text{total}} = \mathbf{1\,\Omega}.$$

Template 3: The 4 Essential Chemical Salts (4 to 5 Marks — Q38/Q41)

Problem:

> Write the chemical formula, preparation equation, and two important uses of:
> (i) Bleaching powder
> (ii) Plaster of Paris

Complete Model Answer:

1. **Bleaching Powder**:
2. **Chemical Formula**: $\mathrm{CaOCl_2}$ (Calcium oxychloride)
3. **Preparation Equation**: Produced by the action of dry chlorine gas on dry slaked lime:

$$\mathrm{Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O}$$

4. **Two Uses**:
   1. As a disinfectant in drinking water to kill germs and make it potable.
   2. For bleaching cotton and linen in the textile industry.
5. **Plaster of Paris (POP)**:
6. **Chemical Formula**: $\mathrm{CaSO_4 \cdot \frac{1}{2}H_2O}$ (Calcium sulphate hemihydrate)

7. **Preparation Equation**: Prepared by heating gypsum ($\mathrm{CaSO_4 \cdot 2H_2O}$) at $373\ \mathrm{K}$ ($100^\circ\mathrm{C}$):

$$\mathrm{CaSO_4 \cdot 2H_2O} \xrightarrow{373\ \mathrm{K}} \mathrm{CaSO_4 \cdot \frac{1}{2}H_2O} + 1\frac{1}{2}\mathrm{H_2O}$$

8. **Two Uses**:
    1. Used by doctors as plaster for supporting fractured bones in the right position.
    2. Used for making statues, decorative casts, and smooth false ceilings.

Template 4: Structure of Nephron & Urine Formation (4 to 5 Marks — Q42)

Problem:

> Describe the structure of a nephron with a labeled diagram. Explain the three main steps involved in urine formation.

Complete Model Answer:

1. **Structure of Nephron**:
2. The nephron is the basic structural and functional filtration unit of the human kidney.

3. It consists of:

    - **Bowman's Capsule**: A cup-like sac at the beginning of a nephron.
    - **Glomerulus**: A dense network (tuft) of blood capillaries enclosed inside Bowman's capsule.
    - **Renal Tubule**: Convoluted tubule divided into proximal tubule, Loop of Henle, and distal tubule.
    - **Collecting Duct**: Collects filtered urine from several nephrons and carries it to the ureter.

4. **Three Steps of Urine Formation**:

5. **Step 1: Ultrafiltration (Glomerular Filtration)**:
    - Blood enters the glomerulus under high pressure. Water, glucose, amino acids, urea, and salts filter across the glomerulus into Bowman's capsule as the initial filtrate. Blood cells and large proteins remain in the blood.
6. **Step 2: Selective Reabsorption**:

    - As the filtrate flows along the renal tubule, essential substances like all glucose, amino acids, major amounts of water, and vital salts are selectively reabsorbed back into surrounding blood capillaries.
7. **Step 3: Tubular Secretion**:
    - Waste substances such as excess potassium ions, hydrogen ions, and drugs are secreted from the blood into the tubule fluid, forming the final urine that enters the collecting duct.

Template 5: Mendel's Monohybrid Cross & Sex Determination (4 to 5 Marks — Q40)

Problem:

> (a) Illustrate Mendel's monohybrid cross between a pure tall pea plant (TT) and a pure dwarf pea plant (tt). State the phenotypic and genotypic ratios of the F_2 generation.
> (b) Explain how the sex of a child is determined in human beings.

Complete Model Answer:

1. **Mendel's Monohybrid Cross**:
2. **Parents**: Pure Tall (TT) $\times$ Pure Dwarf (tt)
3. **Gametes**: T and t
4. F_1 **Generation**: All plants are Tt (All Tall, showing law of dominance).
5. **Selfing** $F_1 \times F_1$ **(**$Tt \times Tt$**)**:

Gametes	T	t
T	TT (Tall)	Tt (Tall)
t	Tt (Tall)	tt (Dwarf)

- F_2 **Phenotypic Ratio**: $3:1$ (3 Tall : 1 Dwarf)
- F_2 **Genotypic Ratio**: $1:2:1$ ($1\,TT:2\,Tt:1\,tt$)
- **Sex Determination in Humans**:
- Human cells contain 23 pairs of chromosomes. Pair 23 are the sex chromosomes.

- Females have two identical sex chromosomes: XX. All female eggs carry an X chromosome ($22 + X$).
- Males have two different sex chromosomes: XY. Half of the male sperms carry an X chromosome ($22 + X$), and half carry a Y chromosome ($22 + Y$).
- **Mechanism**:
  - When an X-carrying sperm fertilizes the egg $\rightarrow XX \implies$ **Female baby (Girl)**.
  - When a Y-carrying sperm fertilizes the egg $\rightarrow XY \implies$ **Male baby (Boy)**.
- **Conclusion**: There is an exact 50% probability of a male or female child, and the genetic sex of the child is determined exclusively by the father's sperm.

Template 6: Cleansing Action of Soap & Micelle Formation (3 to 4 Marks — Q37)

Problem:

> Explain the mechanism of the cleansing action of soaps. What is a micelle?

Complete Model Answer:

1. **Structure of a Soap Molecule**:
2. A soap molecule is a sodium or potassium salt of long-chain carboxylic acid (e.g., sodium stearate $\mathrm{C_{17}H_{35}COONa}$).

3. It has two chemically distinct parts:

    1. **Ionic Head ($-\mathrm{COO^-Na^+}$)**: It is polar and **hydrophilic** (water-loving). It dissolves in water.
    2. **Hydrocarbon Tail ($\mathrm{C_{17}H_{35}-}$)**: It is non-polar and **hydrophobic** (water-fearing). It dissolves in oil, grease, and dirt.

4. **Micelle Formation & Cleansing Action**:

5. Most dirt and grease on clothes are oily in nature and do not dissolve in water.
6. When soap is dissolved in water, soap molecules arrange themselves radially in spherical clusters called **Micelles**:
    - The hydrophobic tails face inwards, dissolving into the central oil/dirt droplet.
    - The hydrophilic ionic heads face outward towards the surrounding water.

7. When the cloth is scrubbed or agitated, the micelles remain suspended as an emulsion due to ion-ion repulsion between heads. The trapped dirt droplet is pulled out into the water and easily rinsed away.

NIOS Class 10 (212) Science & Technology — Chapter-by-Chapter Fast Revision

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Class 10 (212) Science & Technology — Chapter-by-Chapter Fast Revision

> **Only the 19 Public Examination Chapters are summarized here.**
> The 13 TMA chapters have been excluded so your study time is 100% focused on what will be tested.

MODULE 1: MEASUREMENT IN SCIENCE

Lesson 1: Measurement in Science & Technology

- **Fundamental Quantities & SI Units**: Length (m), Mass (kg), Time (s), Temperature (K), Electric Current (A), Luminous Intensity (cd), Amount of Substance (mol).
- **Derived Quantities**: Area (m^2), Volume (m^3), Density ($\mathrm{kg/m}^3$), Speed ($\mathrm{m/s}$), Force (N).
- **Key Instruments**: Vernier Caliper (least count $0.01\ \mathrm{cm} = 0.1\ \mathrm{mm}$), Screw Gauge / Micrometer (least count $0.01\ \mathrm{mm}$).

MODULE 2: MATTER IN OUR SURROUNDINGS (CHEMISTRY)

Lesson 5: Atomic Structure

- **Subatomic Particles**: Electron (-1, Thomson), Proton ($+1$, Rutherford), Neutron (0, Chadwick).
- **Bohr's Model**: Electrons revolve in discrete circular orbits (energy shells K, L, M, N with maximum $2n^2$ electrons: $2, 8, 18, 32$).
- **Atomic Number (Z)** = Number of protons.

- **Mass Number (A)** = Protons + Neutrons.
- **Isotopes vs Isobars**:
- *Isotopes*: Same atomic number Z, different mass number A (e.g. ${}^{12}_{6}\mathrm{C}$ and ${}^{14}_{6}\mathrm{C}$; ${}^{1}_{1}\mathrm{H}$, ${}^{2}_{1}\mathrm{H}$, ${}^{3}_{1}\mathrm{H}$).
- *Isobars*: Same mass number A, different atomic number Z (e.g. ${}^{40}_{18}\mathrm{Ar}$ and ${}^{40}_{20}\mathrm{Ca}$).

Lesson 6: Periodic Classification of Elements

- **Modern Periodic Law**: Physical and chemical properties of elements are periodic functions of their **atomic numbers**.
- **Modern Periodic Table Layout**: 18 vertical columns (**Groups**) and 7 horizontal rows (**Periods**).
- **Periodic Trends**:
- *Valency*: Across period: increases from 1 to 4, then decreases to 0. Down a group: remains constant.
- *Atomic Size (Radius)*: **Decreases** left to right across a period (due to increasing nuclear charge pulling electrons closer); **Increases** down a group (due to addition of new electron shells).
- *Metallic Character*: Decreases across a period; increases down a group.

Lesson 7: Chemical Bonding

- **Octet Rule**: Atoms gain, lose, or share valence electrons to attain a stable noble gas configuration (8 electrons in outermost shell).
- **Ionic (Electrovalent) Bond**: Formed by complete transfer of electrons from a metal to a non-metal (e.g., $\mathrm{NaCl}, \mathrm{MgCl_2}$). High melting points, soluble in water, conduct electricity in molten/aqueous state.
- **Covalent Bond**: Formed by mutual sharing of electron pairs between non-metals (e.g., $\mathrm{H_2}, \mathrm{O_2}, \mathrm{CH_4}, \mathrm{H_2O}$). Lower melting points, non-conductors of electricity.

Lesson 8: Acids, Bases and Salts

- **Acids**: Taste sour, produce $\mathrm{H^+}$ ions in water, turn blue litmus red.
- **Bases**: Taste bitter, soapy to touch, produce $\mathrm{OH^-}$ ions in water, turn red litmus blue.
- **Salts**: Formed by neutralization: $\mathrm{Acid + Base \rightarrow Salt + Water}$.
- **Master Equations**:
- Bleaching Powder: $\mathrm{Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O}$
- Baking Soda: $\mathrm{NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3}$
- POP: $\mathrm{CaSO_4 \cdot 2H_2O \xrightarrow{373\ K} CaSO_4 \cdot \frac{1}{2}H_2O + 1.5H_2O}$

MODULE 3: MOVING THINGS (PHYSICS)

Lesson 10: Force and Motion

- **Newton's First Law**: A body continues in state of rest or uniform motion unless acted upon by an external unbalanced force (Law of Inertia).
- **Newton's Second Law**: Rate of change of momentum is directly proportional to applied force: $F = ma$.
- **Newton's Third Law**: To every action, there is an equal and opposite reaction acting on two different bodies.
- **Law of Conservation of Momentum**: In an isolated system, total momentum before collision equals total momentum after collision: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

Lesson 11: Gravitation

- **Universal Law of Gravitation**: $F = G\frac{Mm}{d^2}$ ($G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$).
- **Acceleration due to Gravity (g)**: $g = \frac{GM}{R^2} \approx 9.8\ \mathrm{m/s^2}$ on Earth; $g_{\mathrm{moon}} = \frac{1}{6}g_{\mathrm{earth}}$.
- **Difference between Mass and Weight**:
- *Mass*: Quantity of matter; constant everywhere; scalar; measured in kg.
- *Weight*: Gravitational force $W = mg$; varies with location; vector; measured in N.
- **Archimedes' Principle**: An immersed body experiences an upward buoyant force equal to the weight of fluid displaced.

MODULE 4: ENERGY (PHYSICS)

Lesson 13: Work and Energy

- **Work**: $W = F \cdot s$ (Joules). No work is done if force is perpendicular to displacement ($\cos 90^\circ = 0$).
- **Kinetic Energy**: $KE = \frac{1}{2}mv^2$.
- **Potential Energy**: $PE = mgh$.
- **Law of Conservation of Energy**: Energy can neither be created nor destroyed; only converted from one form to another. Total mechanical energy $E = KE + PE$ remains constant.
- **Power**: $P = \frac{W}{t}$ (Watts). $1\ \mathrm{kW} = 1000\ \mathrm{W}$.

Lesson 15: Light Energy

- **Reflection**: Angle of incidence $=$ Angle of reflection ($\angle i = \angle r$).
- **Mirror Formula**: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$, Magnification $m = -\frac{v}{u}$.
- **Refraction & Snell's Law**: $\mu = \frac{\sin i}{\sin r}$. Light bends towards normal entering a denser medium.
- **Lens Formula**: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$, Magnification $m = +\frac{v}{u}$.
- **Power of Lens**: $P = \frac{1}{f\,(\mathrm{m})}$ (Dioptres). Convex is $+$, Concave is $-$.
- **Eye Defects**: Myopia (corrected by concave lens) vs Hypermetropia (corrected by convex lens).

Lesson 16: Electrical Energy

- **Ohm's Law**: $V = IR$.
- **Resistance Factors**: $R = \rho\frac{l}{A}$.
- **Resistors**: Series $R_s = R_1 + R_2$; Parallel $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$.
- **Joule's Heating**: $H = I^2Rt$.
- **Commercial Energy**: $1\ \mathrm{kWh} = 3.6 \times 10^6\ \mathrm{J} = 1\ \mathrm{Unit}$.

Lesson 17: Magnetic Effect of Electric Current

- **Right-Hand Thumb Rule**: Thumb indicates current direction; curled fingers indicate circular magnetic field lines.
- **Solenoid**: Coil of insulated wire; magnetic field inside a long current-carrying solenoid is uniform and strong (behaves like a bar magnet).
- **Fleming's Left-Hand Rule**: Used in electric motors (Thumb = Motion, Index = Field, Middle = Current).
- **Safety Devices**: Earth wire (diverts leakage current safely to ground), Fuse (melts and breaks circuit during overload/short-circuit).

MODULE 5: THE LIVING WORLD (BIOLOGY)

Lesson 22: Life Processes – I (Nutrition, Respiration, Circulation, Excretion)

- **Photosynthesis**: Conversion of solar energy to chemical energy in green plants using chlorophyll.
- **Human Digestion**: Salivary amylase in mouth, Pepsin & HCl in stomach, Bile & Pancreatic juice (trypsin, lipase) in small intestine, absorption through villi.

- **Respiration**: Breakdown of glucose: Aerobic (mitochondria, 38 ATP) vs Anaerobic (yeast: ethanol; muscle: lactic acid, 2 ATP).
- **Circulation**: 4-chambered human heart prevents mixing of oxygenated and deoxygenated blood; Double circulation.
- **Excretion**: Nephron filters blood in kidneys; 3 steps: Ultrafiltration $\rightarrow$ Selective Reabsorption $\rightarrow$ Tubular Secretion.

Lesson 23: Life Processes – II (Control & Coordination)

- **Neuron Structure**: Dendrite $\rightarrow$ Cyton (cell body) $\rightarrow$ Axon $\rightarrow$ Nerve ending $\rightarrow$ Synapse (chemical transmission via neurotransmitter).
- **Reflex Arc**: Receptor $\rightarrow$ Sensory neuron $\rightarrow$ Spinal cord $\rightarrow$ Motor neuron $\rightarrow$ Effector muscle.
- **Plant Hormones (Phytohormones)**:
- *Auxin*: Shoot elongation and phototropism.
- *Gibberellin*: Stem growth.
- *Cytokinin*: Promotes cell division.
- *Abscisic Acid (ABA)*: Growth inhibitor, wilting of leaves.
- **Endocrine Glands**: Pancreas (Insulin), Thyroid (Thyroxine requires iodine), Adrenal (Adrenaline - fight/flight hormone).

Lesson 24: Life Processes – III (Reproduction)

- **Asexual Reproduction**: Binary fission (Amoeba), Budding (Yeast, Hydra), Spore formation (Rhizopus), Fragmentation (Spirogyra), Vegetative propagation (Potato, Rose).
- **Sexual Reproduction in Plants**: Flower parts (Stamen = male: anther + filament; Carpel = female: stigma + style + ovary). Pollination (self/cross) followed by fertilization.
- **Human Reproduction**: Testes produce sperm and testosterone; Ovaries produce ova, estrogen, and progesterone. Fertilization occurs in Fallopian tube.

Lesson 25: Heredity

- **Mendel's Laws**: Law of Segregation and Law of Independent Assortment.
- **Monohybrid Cross**: Phenotypic ratio $3:1$, Genotypic ratio $1:2:1$.
- **Sex Determination in Humans**: $44+XX$ (female) and $44+XY$ (male). Father determines child's sex.

MODULE 6: NATURAL RESOURCES (CHEMISTRY)

Lesson 27: Metals and Non-metals

- **Reactivity Series**: $\mathrm{K} > \mathrm{Na} > \mathrm{Ca} > \cdots > \mathrm{Cu} > \mathrm{Ag} > \mathrm{Au}$.
- **Extraction of Metals**:
- Sulphide ores $\rightarrow$ Roasting (excess air).
- Carbonate ores $\rightarrow$ Calcination (limited air).
- **Corrosion**: Oxidation of metals exposed to moisture and air. Prevention: Galvanization (coating iron with zinc), painting, greasing, making alloys (e.g. Stainless steel: $\mathrm{Fe} + \mathrm{Cr} + \mathrm{Ni}$).

Lesson 28: Carbon and Its Compounds

- **Catenation & Tetravalency**: Give rise to huge variety of carbon compounds.
- **Homologous Series**: Family of organic compounds having same functional group and general formula; successive members differ by $-\mathrm{CH_2}-$ unit and $14\ \mathrm{u}$ molecular mass.
- **Soaps and Detergents**: Soaps form micelles to remove grease; ineffective in hard water due to scum formation ($\mathrm{Ca}^{2+}/\mathrm{Mg}^{2+}$ precipitates). Synthetic detergents work effectively even in hard water.

MODULE 7: HUMAN AND ENVIRONMENT (ECOLOGY)

Lesson 29: Natural Environment

- **Ecosystem**: Functional unit comprising biotic components (plants, animals, decomposers) and abiotic components (air, water, soil, sunlight).
- **Food Chain & Food Web**: Linear flow of energy vs interconnected network of food chains.
- **10% Law of Energy**: Only 10% of energy transfers to the next trophic level; limits food chains to 3–4 trophic levels.

Lesson 30: Human Impact on Environment

- **Ozone Layer Depletion**: $\mathrm{O_3}$ layer in stratosphere shields Earth from UV rays. Depleted by CFCs from cooling appliances.
- **Biomagnification**: Progressive accumulation of toxic, non-biodegradable chemicals (like DDT) along food chains.

- **Waste Management**: Segregation into Biodegradable (organic, decomposes naturally) and Non-biodegradable (plastics, glass; requires 3Rs: Reduce, Reuse, Recycle).

NIOS Secondary (Class 10) Science & Technology (212)

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

NIOS Secondary (Class 10) Science & Technology (212)

High-Probability Predicted Mock Examination (85 Marks)

Time: 2½ Hours | Maximum Marks: 85

> **Instructions for the Student**:
> Attempt this paper on **Day 7** in exam-like conditions.
> Time yourself for 2.5 hours. Check your answers against the complete step-by-step solutions at the end.

SECTION A (Objective Type Questions — 42 Marks)

Q1 to Q17: Multiple Choice Questions (1 Mark Each)

1. The SI unit of electric resistance is:
   (A) Volt
   (B) Ampere
   (C) Ohm
   (D) Joule

2. The power of a convex lens of focal length $50\ \text{cm}$ is:
   (A) $+2.0\ \text{D}$
   (B) $-2.0\ \text{D}$
   (C) $+0.5\ \text{D}$
   (D) $+5.0\ \text{D}$

3. A body of mass $5\ \text{kg}$ is moving with an acceleration of $4\ \text{m/s}^2$. The net force acting on it is:
   (A) $1.25\ \text{N}$

(B) $9\ \mathrm{N}$
(C) $20\ \mathrm{N}$
(D) $0.8\ \mathrm{N}$

4. The acceleration due to gravity on the Moon is:
   (A) Equal to that on Earth
   (B) $\frac{1}{6}^{\text{th}}$ of that on Earth
   (C) 6 times that on Earth
   (D) Zero

5. Which defect of vision is corrected by using a concave lens?
   (A) Hypermetropia
   (B) Myopia
   (C) Presbyopia
   (D) Astigmatism

6. Three resistors of $2\ \Omega$ each are connected in series. Their equivalent resistance is:
   (A) $\frac{2}{3}\ \Omega$
   (B) $2\ \Omega$
   (C) $4\ \Omega$
   (D) $6\ \Omega$

7. The chemical formula of Bleaching Powder is:
   (A) CaO
   (B) $\mathrm{CaOCl_2}$
   (C) $\mathrm{Ca(OH)_2}$
   (D) $\mathrm{CaCl_2}$

8. Which of the following substances has a pH greater than 7?
   (A) Lemon juice
   (B) Gastric juice
   (C) Milk of Magnesia
   (D) Pure water

9. Which metal is stored under kerosene oil to prevent spontaneous reaction with air?
   (A) Iron
   (B) Magnesium
   (C) Sodium
   (D) Copper

10. The non-metal which is a good conductor of electricity is:
    (A) Diamond
    (B) Graphite
    (C) Sulphur
    (D) Phosphorus

11. The general molecular formula for Alkanes is:
    (A) $\mathrm{C}_n\mathrm{H}_{2n}$
    (B) $\mathrm{C}_n\mathrm{H}_{2n+2}$
    (C) $\mathrm{C}_n\mathrm{H}_{2n-2}$
    (D) $\mathrm{C}_n\mathrm{H}_{2n+1}\mathrm{OH}$

12. The functional group present in ethanoic acid ($\mathrm{CH_3COOH}$) is:
    (A) $-\mathrm{OH}$
    (B) $-\mathrm{CHO}$
    (C) $-\mathrm{COOH}$
    (D) $>\mathrm{C}=\mathrm{O}$

13. The structural and functional filtration unit of the human kidney is:
    (A) Neuron
    (B) Nephron
    (C) Alveolus
    (D) Villi

14. During anaerobic respiration in yeast, glucose is converted into:
    (A) Lactic acid + Energy
    (B) Ethanol + $\mathrm{CO_2}$ + Energy
    (C) $\mathrm{CO_2}$ + $\mathrm{H_2O}$ + Energy
    (D) Pyruvate only

15. The phenotypic ratio obtained in the F_2 generation of Mendel's monohybrid cross is:
    (A) $1:2:1$
    (B) $3:1$
    (C) $9:3:3:1$
    (D) $2:1$

16. Which part of the human eye controls the amount of light entering the eye?
    (A) Cornea
    (B) Iris

(C) Retina
(D) Ciliary muscles

17. The primary chemical responsible for stratospheric ozone depletion is:
(A) Carbon dioxide
(B) Methane
(C) Chlorofluorocarbon (CFC)
(D) Sulphur dioxide

Q18 to Q27: Objective Questions (2 Marks Each — 1 Mark Per Sub-Part)

1. **Fill in the blanks**:
(i) The device used to measure electric current in a circuit is called `_____`.
(ii) Work done is zero when the angle between force and displacement is `_____` degrees.

2. **True or False**:
(i) Mass of an object changes from place to place. (`TRUE / FALSE`)
(ii) Action and reaction forces act on the same body. (`TRUE / FALSE`)

3. **Fill in the blanks**:
(i) An electric fuse wire has high resistance and a `_____` melting point.
(ii) The commercial unit of electrical energy ($1\ \mathrm{kWh}$) is equal to `_____` Joules.

4. **True or False**:
(i) A concave mirror always forms an erect and virtual image. (`TRUE / FALSE`)
(ii) Violet light suffers maximum deviation when passed through a glass prism. (`TRUE / FALSE`)

5. **Fill in the blanks**:
(i) The chemical formula of Baking Soda is `_____`.
(ii) Plaster of Paris is obtained by heating Gypsum at a temperature of `_____` K.

6. **True or False**:
(i) Pure gold is 24 carats and is too soft for making jewelry. (`TRUE / FALSE`)
(ii) Roasting is the process of heating carbonate ores in limited air. (`TRUE / FALSE`)

7. **Fill in the blanks**:
(i) The unique ability of carbon atoms to form long chains with other carbon atoms is called `_____`.
(ii) Saponification is the chemical reaction used to prepare `_____`.

8. **True or False**:
   (i) Deoxygenated blood from the body tissues enters the left atrium of the human heart. (`TRUE / FALSE`)
   (ii) Insulin is secreted by the thyroid gland. (`TRUE / FALSE`)

9. **Fill in the blanks**:
   (i) In human beings, the normal sex chromosome complement of a female is `_____` .
   (ii) Fertilization of the human egg occurs in the `_____` .

10. **Match the following**:
Column I (Indicator / Phenomenon)	Column II (Observation / Effect)
(i) Phenolphthalein in base	(a) Pink colour
(ii) DDT accumulation in food chain	(b) Biomagnification

Q28: Case Study Question (5 Marks — 1 Mark Per Sub-Part)

1. **Case Study**: A student sets up an electric circuit consisting of an electric heater rated $1000\ \mathrm{W}$, an electric bulb rated $100\ \mathrm{W}$, and a television rated $100\ \mathrm{W}$ operating on a $220\ \mathrm{V}$ domestic power line.
   (i) In which combination (series or parallel) are domestic household appliances connected?
   (ii) Calculate the total electric power consumed when all three appliances are switched on together.
   (iii) What is the total electric current drawn from the $220\ \mathrm{V}$ line?
   (iv) If these appliances run for 5 hours daily, calculate the electrical energy consumed in kWh (units) per day.
   (v) Name the safety wire in domestic circuits that protects users from electric shocks.

SECTION B (Subjective Questions — 43 Marks)

Q29 to Q37: Very Short Answer Questions (2 Marks Each — 18 Marks)

1. State Newton's Second Law of Motion and write its mathematical formula.
2. State the difference between Mass and Weight (any two differences).
3. An object is placed at a distance of $30\ \mathrm{cm}$ in front of a concave mirror of focal length $15\ \mathrm{cm}$. Find the image distance (v).

4. Define 1 Dioptre of power of a lens. What type of lens has a power of $-1.5\ \mathrm{D}$?
5. Write the balanced chemical equation for the preparation of Bleaching Powder from slaked lime. State one of its uses.
6. Why is sodium metal kept immersed in kerosene oil?
7. What is an homologous series? Give the names and molecular formulas of the first two members of the alkane series.
8. Draw a neat labeled diagram of a nerve cell (neuron).
9. State the 10% law of energy transfer in an ecosystem.

Q38 to Q42: Short Answer Questions (3 Marks Each — 15 Marks)

1. Describe the three pathways of glucose breakdown during cellular respiration in:
   (i) Yeast (anaerobic)
   (ii) Human muscle cells (lack of oxygen)
   (iii) Mitochondria (aerobic)
2. Write the chemical formula, preparation equation, and two important uses of Plaster of Paris (POP).
3. Explain Mendel's monohybrid cross between a pure tall pea plant (TT) and a pure dwarf pea plant (tt). Show the Punnett square for the F_2 generation and state its phenotypic ratio.
4. (a) Define electric power.
   (b) An electric iron of resistance $20\ \Omega$ draws a current of $5\ \mathrm{A}$. Calculate the heat developed in 30 seconds.
5. Differentiate between Roasting and Calcination with one balanced chemical equation for each.

Q43 to Q44: Long Answer Questions (5 Marks Each — 10 Marks)

1. (a) What is Myopia? State its two causes.
   (b) Draw neat ray diagrams showing:
   (i) a myopic eye, and
   (ii) correction of myopia using an appropriate lens.
   (OR)
   (a) State Ohm's law. Draw a schematic circuit diagram for its verification in the laboratory.
   (b) Three resistors of resistances $3\ \Omega, 6\ \Omega$, and $2\ \Omega$ are given. How would you connect them to obtain an equivalent resistance of $4\ \Omega$? Show the calculation.

2. (a) Describe the structure of a nephron with a labeled diagram.
   (b) Briefly explain the three main steps of urine formation in the human kidney.

(OR)

(a) Explain why carbon forms a vast number of compounds (explain catenation and tetravalency).

(b) Explain the cleansing action of soaps with the help of a labeled diagram of a micelle.

COMPLETE STEP-BY-STEP SOLUTIONS & MARKING SCHEME

Section A Solutions

- **Q1**: (C) Ohm (Ω)
- **Q2**: (A) $+2.0\ \text{D}$ [$P = \frac{1}{0.5\ \text{m}} = +2.0\ \text{D}$]
- **Q3**: (C) $20\ \text{N}$ [$F = ma = 5 \times 4 = 20\ \text{N}$]
- **Q4**: (B) $\frac{1}{6}^{\text{th}}$ of that on Earth
- **Q5**: (B) Myopia
- **Q6**: (D) $6\ \Omega$ [$R_s = 2 + 2 + 2 = 6\ \Omega$]
- **Q7**: (B) CaOCl_2
- **Q8**: (C) Milk of Magnesia [pH ≈ 10]
- **Q9**: (C) Plaster of Paris
- **Q10**: (C) Strongly acidic
- **Q11**: (B) $\text{C}_n\text{H}_{2n+2}$
- **Q12**: (C) $-\text{COOH}$ (Carboxylic acid)
- **Q13**: (B) Nephron
- **Q14**: (B) Ethanol + CO_2 + Energy
- **Q15**: (B) $3:1$
- **Q16**: (B) Iris
- **Q17**: (C) Chlorofluorocarbon (CFC)
- **Q18**: (i) ammeter | (ii) 90° [$W = Fs\cos 90^\circ = 0$]
- **Q19**: (i) FALSE [Mass is constant] | (ii) FALSE [Act on different bodies]
- **Q20**: (i) low | (ii) $3.6 \times 10^6\ \text{J}$
- **Q21**: (i) FALSE [Convex mirror is always virtual/erect] | (ii) TRUE
- **Q22**: (i) NaHCO_3 | (ii) $373\ \text{K}$ (100°C)
- **Q23**: (i) TRUE | (ii) FALSE [Heating carbonate ore is Calcination]
- **Q24**: (i) catenation | (ii) soap
- **Q25**: (i) FALSE [Enters right atrium] | (ii) FALSE [Secreted by pancreas]
- **Q26**: (i) XX | (ii) Fallopian tube (Oviduct)

- **Q27**: (i) $\rightarrow$ (a) Pink colour | (ii) $\rightarrow$ (b) Biomagnification
- **Q28**: (i) Parallel combination. (ii) Total power $= 1000 + 100 + 100 = \mathbf{1200}\ \mathrm{W} = 1.2\ \mathrm{kW}$. (iii) Current $I = \frac{P}{V} = \frac{1200}{220} \approx \mathbf{5.45}\ \mathrm{A}$. (iv) Energy $= 1.2\ \mathrm{kW} \times 5\ \mathrm{h} = \mathbf{6}\ \mathrm{kWh}\ (\text{units})$. (v) Earth wire (Green/Yellow).

Section B Detailed Solutions

Q29:
Newton's Second Law: The rate of change of momentum of a body is directly proportional to the applied unbalanced force and takes place in the direction of the force.
Formula: $F = ma = \frac{m(v-u)}{t}$.

Q30:
1. *Mass*: Quantity of matter in a body; constant everywhere; scalar unit kg.
2. *Weight*: Gravitational pull $W = mg$; varies from place to place; vector unit N.

Q31:
$u = -30\ \mathrm{cm}, f = -15\ \mathrm{cm}$.
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \implies \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-15} - \frac{1}{-30} = -\frac{1}{15} + \frac{1}{30} = -\frac{1}{30}$.
$\mathbf{v = -30}\ \mathrm{cm}$. The image is real, inverted, and formed at the centre of curvature.

Q32:
1 Dioptre is the power of a lens whose focal length is 1 metre ($1\ \mathrm{D} = 1\ \mathrm{m}^{-1}$).
A power of $-1.5\ \mathrm{D}$ indicates a **Concave Lens** (diverging lens).

Q33:
$\mathrm{Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O}$.
Use: As a disinfectant for water purification or bleaching agent in textile mills.

Q34:
Sodium is an extremely reactive alkali metal. It reacts vigorously with atmospheric oxygen and water vapour, producing hydrogen gas and catching fire. To prevent this, it is stored submerged under kerosene oil.

Q35:
An homologous series is a group of organic compounds sharing the same functional group and general formula, where consecutive members differ by a $-\mathrm{CH_2}-$ unit ($14\ \mathrm{u}$).
First two alkanes: Methane ($\mathrm{CH_4}$) and Ethane ($\mathrm{C_2H_6}$).

Q36:

Neuron diagram consists of:

- Dendrites (receive signals)
- Cyton / Cell body (contains nucleus)
- Axon (transmits nerve impulses)
- Nerve endings (synaptic transmission).

Q37:

10% Law: During energy transfer across trophic levels in an ecosystem, only about 10% of the energy is stored as biomass in the next level; the remaining 90% is dissipated as heat or used in respiration.

Q38:

Glucose ($6\mathrm{C}$) breaks down in cytoplasm into Pyruvate ($3\mathrm{C}$):

1. *In Yeast (Absence of $\mathrm{O_2}$)*: Ethanol + $\mathrm{CO_2}$ + $2\ \mathrm{ATP}$.
2. *In Muscle Cells (Lack of $\mathrm{O_2}$)*: Lactic acid + Energy (causes muscle fatigue).
3. *In Mitochondria (Presence of $\mathrm{O_2}$)*: $\mathrm{CO_2} + \mathrm{H_2O} + 38\ \mathrm{ATP}$.

Q39:

Formula: $\mathrm{CaSO_4 \cdot \frac{1}{2}H_2O}$.

Preparation: $\mathrm{CaSO_4 \cdot 2H_2O} \xrightarrow{373\ \mathrm{K}} \mathrm{CaSO_4 \cdot \frac{1}{2}H_2O} + 1\tfrac{1}{2}\mathrm{H_2O}$.

Uses: 1. Bone fracture casts. 2. Statues and false ceilings.

Q40:

Cross: $TT \times tt \rightarrow F_1 : Tt$ (All Tall).

Selfing: $Tt \times Tt \rightarrow F_2 : TT, Tt, Tt, tt$.

Phenotypic ratio: $3:1$ (3 Tall : 1 Dwarf). Genotypic ratio: $1:2:1$.

Q41:

(a) Electric power is the rate at which electrical energy is consumed in an electric circuit: $P = VI = I^2R$.

(b) Heat $H = I^2Rt = (5)^2 \times 20 \times 30 = 25 \times 600 = \mathbf{15,000}$ Joules ($15\ \mathrm{kJ}$).

Q42:

1. *Roasting*: Heating sulphide ores in excess air:

$2\mathrm{ZnS} + 3\mathrm{O_2} \xrightarrow{\Delta} 2\mathrm{ZnO} + 2\mathrm{SO_2}$.

2. *Calcination*: Heating carbonate ores in limited/no air:

$\mathrm{ZnCO_3} \xrightarrow{\Delta} \mathrm{ZnO} + \mathrm{CO_2}$.

Q43:

(a) Myopia is short-sightedness. Causes: excessive curvature of lens, elongation of eyeball.

(b) Ray diagrams:

- Defective eye: rays converge in front of retina.
- Corrected eye: concave lens diverges rays to focus sharply on retina.

(OR)

(a) $V = IR$. Circuit diagram with battery, key, ammeter, voltmeter, resistor, rheostat.

(b) Connect $3\,\Omega$ and $6\,\Omega$ in parallel: $\frac{1}{R_p} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2} \implies R_p = 2\,\Omega$.

Then connect with $2\,\Omega$ in series: $R_{\text{total}} = 2 + 2 = \mathbf{4\,\Omega}$.

Q44:

(a) Nephron: Bowman's capsule, Glomerulus, Convoluted tubule, Collecting duct.

(b) 1. Ultrafiltration in glomerulus $\rightarrow$ 2. Selective reabsorption of glucose, water, amino acids $\rightarrow$ 3. Tubular secretion of wastes.

(OR)

(a) Catenation (carbon bonds with carbon to form chains/rings) and Tetravalency (valency 4 allows bonding with 4 atoms).

(b) Soap molecule has hydrophilic head ($-\mathrm{COO^-Na^+}$) and hydrophobic tail (hydrocarbon). In water, tails dissolve in central grease droplet while heads face outward, forming a spherical **Micelle** that lifts dirt into water.